

TraceMask — Presentation Script

Kalpvruksh 2.0 Mini Hackathon · Problem P16 · Team Cryptic_Vruksh (T049) ·
5-minute pitch + optional 1-minute demo



The one idea to leave with the judges: Every website gets its own email address. If a stranger ever uses it, we know exactly which website leaked it, and we give you the proof.

Timing plan

#	SLIDE	TIME	LENGTH
1	Title page	0:00 – 0:20	20 sec
2	Idea: TraceMask	0:20 – 1:40	80 sec
3	Technical approach	1:40 – 2:40	60 sec
4	Feasibility and viability	2:40 – 3:30	50 sec
5	Impact and benefits	3:30 – 4:20	50 sec
6	Research and references	4:20 – 5:00	40 sec
+	Live demo (optional)	5:00 – 6:00	60 sec

Delivery tips

- Practise with a timer. The full script is 5 minutes; keep 1 extra minute for the demo.
- Speak slowly. Pause after every number (24/24, 2 seconds, 13.73 crore).
- Point at the slide area you are talking about. Each section below says where.
- Use "you" and "your email". Judges should picture their own inbox.
- If you have 2–3 speakers: Speaker A takes slides 1–2, Speaker B slides 3–4, Speaker C slides 5–6. Hand over with "Now, [name] will show how it works."

The story in 30 seconds (use if time is cut)

"We give the same email to every website. When one leaks it, we never know who did it. **TraceMask** checks a website's risk before you sign up, gives that site its own real email address, and watches your inbox. If anyone else ever uses that address, TraceMask names the website that leaked it within seconds, with proof, and writes the data-erasure request for you. It runs on your own laptop, costs nothing, and passed 24 out of 24 security checks."

1

Slide 1: Title page

0:00 – 0:20 · 20 sec



WHAT TO SAY

Good morning, respected judges. We are **Team Cryptic_Vruksh**, Team ID **T049**.

Our problem statement is **P16: Exposure of Personal Identity Through Routine Online Sign-Ups**, in the Cybersecurity domain.

Our solution is called **TraceMask**: a different you for every website, and proof of who leaked it.

On screen: Title slide: problem ID, domain, Team ID T049, team logo.

Remember: Say the product name and the one-line promise clearly. Judges should remember "TraceMask".

2

Slide 2: Idea: TraceMask

0:20 – 1:40 • 80 sec

TRACEMASK: IDENTITY FIREWALL FOR ONLINE SIGN-UPS
Proposed Solution (Describe your Idea/Solution/Prototype)

TraceMask is a local-first web app that gives every sign-up its own real, deliverable email alias, checks each website live before any data is shared, automatically proves which company leaked or sold your identity, and turns that proof into a PDF evidence report and a DPDP/GDPR erasure request.

1. Detailed explanation of the proposed solution	2. How it addresses the problem	3. Innovation and uniqueness of the solution
1. SCAN before sign-up Scan the website on iOS, Android, or a desktop app. TraceMask checks the website's certificate, domain age, privacy policy, and more. 2. MASK with one alias per site Mask your real email with a unique, disposable email alias. The alias is linked to your real email but can't be traced back to it. 3. TRACE leaks automatically After you sign up, TraceMask automatically checks if your email is leaked. If it is, it shows you the proof and the company that leaked it. 4. ACT on the evidence Use the evidence to demand a PDF evidence report and a DPDP/GDPR erasure request.	P16 pain point A real address is a pain point. It's hard to keep track of all the addresses you use. It's also hard to know if your address is leaked. TraceMask answer TraceMask gives you a unique, disposable email alias for each website. This way, you can keep track of all your addresses and know if your address is leaked. How it works TraceMask automatically checks if your email is leaked. If it is, it shows you the proof and the company that leaked it. You can then use this evidence to demand a PDF evidence report and a DPDP/GDPR erasure request.	Leak verification, not just blocking TraceMask doesn't just block websites that are leaking your data. It also verifies if the website is leaking your data. This way, you can know if your data is leaked and take action. Risk-aware identity TraceMask gives you a unique, disposable email alias for each website. This way, you can keep track of all your addresses and know if your address is leaked. Evidence-grade output TraceMask gives you a PDF evidence report and a DPDP/GDPR erasure request. This way, you can have proof of your identity and demand a PDF evidence report and a DPDP/GDPR erasure request.

On screen: Start with the problem, then walk through the four coloured steps on the left, then point to the table and the comparison.

WHAT TO SAY

(Start with the problem.) Think about how many websites have your email: shopping apps, job portals, coupon sites, event forms. We give them all the **same email**.

If even one of them leaks or sells it, spam and phishing start, and you can **never find out who did it**. Unsubscribing only stops the mails; your data stays with them. And if you give fake details, password reset stops working.

(Point to the four steps.) TraceMask fixes this in four simple steps.

1. Scan. Before you sign up, you paste the website. TraceMask checks it live: is the certificate valid, how old is the domain, was it breached before, how many trackers it has, does it have a privacy policy. You get a **risk score out of 100**.

2. Mask. Instead of your real email, you get a **unique email address for that one site**, like `yourname+mynta-x7k2p@gmail.com`. It is a real address, so welcome mails and password resets still reach your inbox.

3. Trace. Only that one website knows this address. So if **anyone else** ever emails it, that website leaked it. TraceMask catches this in about **2 seconds** and shows proof from the email's security checks.

4. Act. The leaked mail is moved to quarantine, and you get a **PDF evidence report** and a ready **data-erasure letter** under India's DPDP Act.

(Point to the middle table.) Every pain point in P16 has a direct answer here.

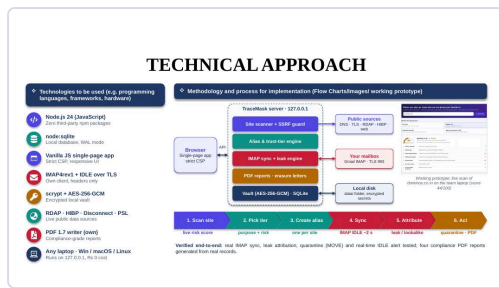
(Point to the right.) What makes us unique: tools like Firefox Relay or SimpleLogin only hide your email, and you have to track leaks by hand. TraceMask **names the company that leaked** automatically, checks risk **before** you sign up, and gives you **evidence**.

Remember: Tell it as a story: same email everywhere → leak → no one to blame → TraceMask gives each site its own address, so the leaker is caught.

3

Slide 3: Technical approach

1:40 – 2:40 • 60 sec



On screen: Left: technology list. Centre: architecture diagram. Right: screenshot of a real scan. Bottom: 6-step flow.

WHAT TO SAY

(Left side.) We built TraceMask with **Node.js** and **plain JavaScript**, with **zero third-party packages**, so there is no supply-chain risk. Data is stored locally in SQLite, and passwords are encrypted with **AES-256**. It runs on any laptop at zero cost.

(Diagram.) The browser talks to the TraceMask server, which runs **only on your own computer**. Inside it:

the **site scanner** checks public sources like domain records and breach databases;

the **alias engine** creates the addresses;

the **leak engine** reads only the email **headers** from your Gmail, never the message itself;

the **report module** makes the PDFs;

and the **vault** keeps secrets encrypted.

(Bottom flow.) The whole process is six steps: scan, pick tier, create alias, sync, attribute, act.

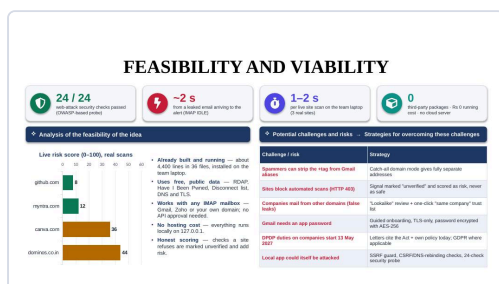
(Screenshot.) This is a **real scan of dominos.co.in** from our laptop. It scored 44 and found a 2021 breach of **2.25 crore accounts**.

Remember: Keep it simple: "Runs on your laptop, reads only headers, keeps secrets encrypted, uses real public data."

4

Slide 4: Feasibility and viability

2:40 – 3:30 • 50 sec



On screen: Top: four number tiles. Left: bar chart of real scores. Right: risks and how we handle them.

WHAT TO SAY

This is not just an idea. It is **built and working**.

(Tiles.) It passed **24 out of 24** security attack checks. A leak alert comes about **2 seconds** after the email arrives. Scanning a website takes **1 to 2 seconds**. And it needs **no paid services and no cloud server**.

(Chart.) These are real scores from our live scans. GitHub scored 8 and Myntra 12, which is low risk. Canva scored 36 and Domino's 44, moderate risk, mainly because of past breaches and trackers.

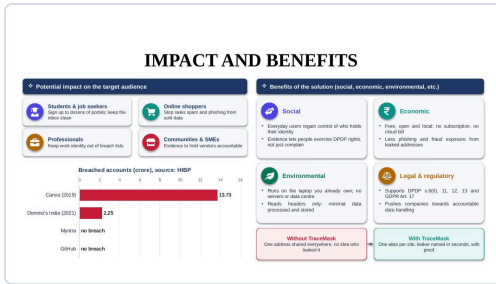
(Table: pick two rows.) We also planned for the risks. Spammers can remove the plus tag from a Gmail alias, so we support **your own domain** as well. Some websites block scanners; we mark that as **unverified** and add risk instead of calling it safe.

Remember: Numbers prove feasibility. Read them slowly: 24/24, 2 seconds, 1-2 seconds, zero cost.

5

Slide 5: Impact and benefits

3:30 – 4:20 • 50 sec



On screen: Left: who benefits and the breach chart. Right: four benefit cards and the before/after boxes.

WHAT TO SAY

(Left top.) Who benefits: **students and job seekers** who sign up everywhere, **online shoppers, professionals, and small communities and businesses.**

(Chart.) Why it matters: Canva's 2019 breach exposed **13.73 crore** accounts, and Domino's India's 2021 breach exposed **2.25 crore**. If you used your real email there, it is out forever. With TraceMask, only that site's alias would be exposed, and you would know exactly who leaked it.

(Cards.) **Socially**, people get control and proof. **Economically**, it is free. **Environmentally**, it runs on your own laptop with no servers. **Legally**, it supports DPDP rights like erasure.

(Bottom boxes.) Without TraceMask: one address everywhere and no idea who leaked it. With TraceMask: one alias per site, and the leaker is named in seconds, with proof.

Remember: End this slide on the before/after line. It is the easiest idea for judges to repeat.

6

Slide 6: Research and references

4:20 – 5:00 • 40 sec



On screen: Left: reference table with links. Right: research findings, DPDP timeline, validation.

WHAT TO SAY

Our work is based on real research: the **DPDP Act 2023** and **DPDP Rules 2025**, email security standards, breach databases, and **OWASP** security guidelines. All are listed here with links.

(Right top.) Our key finding: email alias services already exist, but **none of them automatically detects who leaked your data**. That is the gap TraceMask fills.

(Timeline.) The timing is right. Under the DPDP Rules, company duties, including erasure, apply from **13 May 2027**.

(Close.) **TraceMask: a different you for every website, and proof of who leaked it.** Thank you. We are happy to show you a live demo.

Remember: Pause after "Thank you". Offer the demo; do not rush into it unless the judges agree.

Optional live demo (60 seconds)

1. Open TraceMask → **Protect a sign-up** → type a real site → **Scan site**. Say: "It checks the site live in about a second."
2. Choose **Regular use** → **Create identity** → copy the alias. Say: "This address is only for this site."
3. A teammate sends an email to that alias from a different email account. Click **Sync now** if the alert has not appeared. Say: "Only this site had this address, so this is a leak."
4. Open **Leak Center** → **Evidence report (PDF)** → **Draft erasure request**. Say: "Proof and action in one click."

Before the demo: unlock TraceMask, click **Sync now** once, keep the teammate's second email account open, and have the Leak Center ready in another tab.

Simple meanings of technical words

WORD	SAY IT LIKE THIS
Alias	A separate email address that still delivers to your inbox. Each website gets its own.
Plus-addressing	Gmail delivers <i>you+anything@gmail.com</i> to <i>you@gmail.com</i> . We use it to create aliases without any setup.
Email headers	The "envelope" of an email: sender, receiver, date, subject. TraceMask reads only these, never the message.
IMAP / IDLE	The standard way apps read email. IDLE means Gmail tells us instantly when new mail arrives.
SPF, DKIM, DMARC	Security checks your mail provider runs. They prove which company really sent an email.
Have I Been Pwned	A trusted public list of real data breaches (which site, when, how many accounts).
RDAP	The official record of when a website domain was registered. New domains are riskier.
Trackers	Third-party scripts (ads, analytics) that follow users across websites.
Quarantine	A separate mail folder where leaked or blocked mail is moved. Nothing is deleted.
SHA-256	A digital fingerprint. If even one letter of the evidence changes, the fingerprint changes.
DPDP Act 2023	India's data protection law. It gives people the right to know, correct and erase their data.
127.0.0.1 / local-first	The app runs only on your own computer. Your data never goes to our servers.

Likely judge questions — short answers

Q1. How is this different from Firefox Relay or Apple Hide My Email?

Those tools only hide your address. You still have to work out a leak yourself. TraceMask automatically names the company that leaked, checks website risk before sign-up, and gives you a PDF evidence report and an erasure letter.

Q2. Do you read my emails?

No. We read only the headers (sender, receiver, date, subject and security results). Message bodies are never downloaded.

Q3. What if a company sends mail from a different domain?

If the sender looks like the same brand (e.g. safebank.com vs safebank.in), we mark it "Lookalike: review" instead of calling it a leak. The user can mark it "same company" with one click.

Q4. Can someone fake a leak to blame a company?

We separate the cases. If the sender passes DKIM/DMARC, it is a real company (high confidence). If not, it is shown as medium confidence, typical of spam lists.

Q5. Is my Gmail password safe?

We use a Gmail app password, not the real one. It is encrypted with AES-256-GCM using a key from your master password, which is never stored. Everything stays on your laptop.

Q6. What if I forget the master password?

For security it cannot be recovered. You simply reconnect your mailbox; the aliases themselves keep working in Gmail.

Q7. Does it only work with Gmail?

No. It works with any mailbox that supports IMAP over TLS (Zoho, your own domain). With your own domain, each alias is a completely separate address.

Q8. Is the erasure letter legally valid today?

The DPDP duties on companies apply from 13 May 2027. Until then the letter is a formal request citing the Act and the company's own policy. GDPR applies to EU/UK companies.

Q9. How did you test security?

We ran 24 attack checks based on OWASP. We found and fixed six issues, including a server crash from a malformed link and a server-side request forgery risk in the scanner. All 24 now pass.

Q10. What is next (future scope)?

A browser extension that fills the alias into sign-up forms automatically, phone-number masking, and a community "leak index" of which companies leak most.